

DevOps Project 3: Build Automation Using Maven

Goal

Automate the build process of a Java application using **Maven**.

Clone the Source code : <https://github.com/prade01>

Simple maven-project

Tools and Technology Used

- JDK11
- MAVEN3.8
- Git
- WSL Linux
- VS Code Editor

Commands Used to Automate Maven Project

```
prade@DELL:~$ mvn archetype:generate  
-DgroupId=xyz.shrutideep.project  
-DartifactId=maven-project  
-DarchetypeArtifactId=maven-archetype-quickstart  
-DinteractiveMode=false
```

This creates a standard Maven project structure as
maven-project

```
maven-project/
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   ├── xyz
│   │   │   │   ├── shrutideep
│   │   │   │   │   ├── project
│   │   │   │   │   │   App.java
│   │   └── test
│   │       ├── java
│   │       │   ├── xyz
│   │       │   │   ├── shrutideep
│   │       │   │   │   ├── project
│   │       │   │   │   │   AppTest.java
│   └── target
│       ├── classes
│       │   ├── xyz
│       │   │   ├── shrutideep
│       │   │   │   ├── project
│       │   │   │   │   App.class
│       ├── generated-sources
│       │   └── annotations
│       ├── generated-test-sources
│       │   └── test-annotations
│       ├── maven-archiver
│       │   └── pom.properties
│       ├── maven-project-1.0-SNAPSHOT.jar
│       ├── maven-status
│       │   ├── maven-compiler-plugin
│       │   │   ├── compile
│       │   │   │   ├── default-compile
│       │   │   │   │   ├── createdFiles.lst
│       │   │   │   │   └── inputFiles.lst
│       │   └── testCompile
│       └── testClasses
```

```
├── testCompile
│   ├── default-testCompile
│   │   ├── createdFiles.lst
│   │   └── inputFiles.lst
├── surefire-reports
│   ├── TEST-xyz.shrutideep.project.AppTest.xml
│   └── xyz.shrutideep.project.AppTest.txt
├── test-classes
│   ├── xyz
│   │   ├── shrutideep
│   │   │   ├── project
│   │   │   │   AppTest.class
```

Adding Dependencies in pom.xml

```
pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
2      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4                          http://maven.apache.org/maven-v4_0_0.xsd">
5
6      <modelVersion>4.0.0</modelVersion>
7
8      <groupId>xyz.shrutideep.project</groupId>
9      <artifactId>maven-project</artifactId>
10     <version>1.0-SNAPSHOT</version>
11     <packaging>jar</packaging>
12
13     <name>maven-project</name>
14     <url>http://maven.apache.org</url>
15
16
17     <properties>
18         <maven.compiler.source>8</maven.compiler.source>
19         <maven.compiler.target>8</maven.compiler.target>
20         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
21     </properties>
22
23     <dependencies>
24         <dependency>
25             <groupId>junit</groupId>
26             <artifactId>junit</artifactId>
27             <version>3.8.1</version>
28             <scope>test</scope>
29         </dependency>
30     </dependencies>
31
32     <build>
33         <plugins>
```

```
pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
23  <dependencies>
24  <dependency>
28      <scope>test</scope>
29  </dependency>
30 </dependencies>
31
32 <build>
33 <plugins>
34 <plugin>
35     <groupId>org.apache.maven.plugins</groupId>
36     <artifactId>maven-jar-plugin</artifactId>
37     <version>3.3.0</version>
38     <configuration>
39         <archive>
40             <manifest>
41                 <mainClass>xyz.shrutideep.project.App</mainClass>
42             </manifest>
43         </archive>
44     </configuration>
45 </plugin>
46 </plugins>
47 </build>
48
49 </project>
50
51
```

Build the project using Maven tool

\$cd maven-project/ \$mvn clean install

```

[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ maven-project ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/prade/maven-project/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ maven-project ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /home/prade/maven-project/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ maven-project ---
[INFO] Surefire report directory: /home/prade/maven-project/target/surefire-reports

-----
T E S T S
-----
Running xyz.shrutideep.project.AppTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.017 sec

Results :

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO]
[INFO] --- maven-jar-plugin:3.3.0:jar (default-jar) @ maven-project ---
[INFO] Building jar: /home/prade/maven-project/target/maven-project-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ maven-project ---
[INFO] Installing /home/prade/maven-project/target/maven-project-1.0-SNAPSHOT.jar to /home/prade/.m2/repository/xyz/shrutideep/project/maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.jar
[INFO] Installing /home/prade/maven-project/pom.xml to /home/prade/.m2/repository/xyz/shrutideep/project/maven-project/1.0-SNAPSHOT/maven-project-1.0-SNAPSHOT.pom
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.623 s
[INFO] Finished at: 2026-02-10T01:18:34Z
[INFO]
prade@DELL:~/maven-project$

```

#Post Build Success check for Jar/war files in target/ subdirectory

Test App post build Success

\$java -jar target/*.jar

```

prade@DELL:~/maven-project$ java -jar target/maven-project-1.0-SNAPSHOT.jar
A successfully built Java application using Maven
prade@DELL:~/maven-project$

```